

PRANAY KUMBHARE

AI Validation Tester | GxP & Computer System Validation (CSV) | Pharmaceutical Manufacturing | AI/LLM Quality Engineering

Pune, India | +91 8149120948 | pranaykumbhare1204@gmail.com | 7.8+ Years Experience

PROFESSIONAL SUMMARY

Senior SDET / AI Validation Tester with 7.8+ years of Functional, Automation, and AI/LLM Testing experience, including direct exposure to GxP-regulated pharmaceutical environments. Currently validating a cloud-based AI/Data platform for Takeda Pharmaceutical (OneData Platform) across 4 GxP environments, and previously validated an AI Clinical Content Generation platform for Roche Diagnostics against regulatory and compliance requirements. Skilled in translating business, NFR, and AI/regulatory requirements into test plans, functional and automation test suites, and release/compliance reports within Agile/SAFe teams using JIRA. Experienced in testing AI-generated/LLM outputs for accuracy, groundedness, and hallucination risk (LangSmith) to support regulatory review, alongside REST API, backend/database (MySQL), and performance testing. Comfortable working with GxP documentation practices, computer system validation concepts, and cross-functional pharma stakeholders, and quick to pick up new mandatory tools such as RPA platforms and Microsoft Copilot.

ALIGNMENT TO ROLE REQUIREMENTS

- **Functional Testing:** 7.8+ years of functional, regression, smoke, and end-to-end testing across Healthcare, Pharmaceutical, Payments, Logistics, and Enterprise platforms.
- **GxP & Regulatory Compliance:** Tested AI/data platforms across 4 GxP environments for Takeda Pharmaceutical (OneData Platform) and supported regulatory review of AI-generated clinical content for Roche Diagnostics (AIDCG platform).
- **Pharmaceutical Manufacturing / Domain Exposure:** Direct project experience with pharma data pipelines, ETL workflows, and clinical content generation, working alongside compliance and quality stakeholders in regulated pharma environments.
- **Agile & Tooling:** Active participant in Agile/SAFe ceremonies (PI Planning, Iteration Planning, ART Sync) using JIRA and Confluence for defect and requirement tracking; comfortable adapting to qTest for test management.
- **AI Validation Focus:** Hands-on testing of LLM/Agentic AI outputs using LangSmith for accuracy, groundedness, and hallucination risk against compliance rules — directly applicable to AI validation in a regulated pharma setting.
- **RPA & Microsoft Copilot:** No direct production RPA implementation to date; strong automation-scripting foundation (Playwright, Selenium, Python) that transfers readily to RPA tools, and open to ramping up quickly on RPA and Copilot as needed.

CORE COMPETENCIES

Validation & Compliance: GxP Environments, Computer System Validation (CSV) Awareness, Regulatory Compliance Testing, Audit-Ready Documentation, Risk-Based Test Strategy

AI / LLM Quality Engineering: LLM Output Validation, Contextual Accuracy & Groundedness, Hallucination Risk Assessment, Prompt/Response Verification, Agentic AI Functional Testing

Automation Testing: Playwright (TypeScript), Selenium WebDriver, Page Object Model (POM), Smoke/Regression/End-to-End Testing, Cross-Browser & Cross-Device Testing

Programming & Scripting: TypeScript, JavaScript (ES6+, async/await), Python (Requests, Pandas), SQL

API & Backend Testing: REST API Testing, Postman, Newman, JSON, OAuth/JWT, Microservices Testing, SQL/MySQL Validation

Performance Engineering: Apache JMeter, NeoLoad, BlazeMeter, Load/Stress/Spike/Endurance/Soak Testing, Workload Modeling, NFR Gathering, RCA

Cloud, DevOps & Observability: AWS (S3, IAM, SNS, SQS, KMS, API Gateway, CloudWatch), Harness CI/CD, Jenkins, Git/GitHub, Dynatrace

Delivery & Reporting: Test Strategy & Planning, KPI Dashboards, Defect/Risk Analysis, Release Readiness, Agile/SAFe (PI Planning, ART Sync), JIRA, Stakeholder Management

TECHNICAL SKILLS

Automation: Playwright (TypeScript), Selenium WebDriver, Page Object Model, Regression & Smoke Testing

Programming: TypeScript, JavaScript (ES6+), Python (Requests, Pandas), SQL

API / Backend: REST API, Postman, Newman, JSON, OAuth, JWT, Microservices Testing

Performance: Apache JMeter, NeoLoad, BlazeMeter, Locust, Load/Stress/Endurance/Soak Testing

Database: MySQL, SQL Validation, Data Integrity Validation, ETL Workflow Testing

Cloud: AWS - S3, IAM, SNS, SQS, KMS, API Gateway, CloudWatch, Databricks

Monitoring: Dynatrace, Grafana, CloudWatch

DevOps: Harness CI/CD, Jenkins, Git, GitHub, Newman CLI

AI / LLM Testing: LangSmith, LLM Output Validation, Contextual/Groundedness Checks, Hallucination Risk Assessment, Agentic AI Testing

Validation & PM Tools: JIRA, Confluence, Figma, BrowserStack (qTest — quick learner)

PROFESSIONAL EXPERIENCE

Senior Performance & Automation Engineer

Aug 2025 – Present

Ancile Digital Solution LLP

Client: Takeda Pharmaceutical | Project: OneData Platform | Domain: Pharmaceutical, Cloud Data Platform, AI, Data Engineering

- Led NFR gathering, workload modeling, and test strategy for Takeda's OneData Platform, a cloud data platform with 15+ data pipelines and ETL workflows in a GxP-regulated pharmaceutical environment.
- Built Playwright (TypeScript) automation scripts using the Page Object Model for smoke and regression testing to check application stability ahead of validation and performance testing.
- Automated REST API testing for data ingestion and ETL endpoints using Postman, Newman, and Python (Requests, Pandas), speeding up response checks and compliant test-data setup.
- Ran and analyzed JMeter and NeoLoad performance tests with 200–500 concurrent users; identified 4 throughput bottlenecks and cut average response time by ~35%.
- Tested microservices on AWS (SNS, IAM, KMS, S3, CloudWatch) across 4 GxP environments, validating auto-scaling, load balancer (ALB), and message queue (SQS/Kafka) behavior against compliance expectations.
- Integrated automation and performance scripts into Harness CI/CD pipelines, cutting pipeline failure rate from 18% to under 5%.
- Used LangSmith to validate LLM/Agentic AI features across 3 platform modules, assessing output accuracy, groundedness, and hallucination risk against pharma compliance rules.
- Authored test strategy, execution reports, KPI dashboards, and risk reports; participated in SAFe PI Planning, Iteration Planning, and ART Sync using JIRA.

Client: Roche Diagnostics | Project: AIDCG (AI Data Clinical Content Generation) Platform | Domain: Healthcare, Artificial Intelligence, Clinical Content Generation

- Gathered NFRs and built a combined functional, automation, API, backend, and performance test plan across 8 modules of an AI clinical content platform used in regulated healthcare workflows.
- Built Postman collections of 80+ REST API test cases and Python (Requests) scripts to validate response, payload, and business rules; logged and tracked 30+ defects through JIRA to resolution.
- Validated AI-generated clinical content against 5 source document types for accuracy, groundedness, and hallucination risk to support regulatory review of LLM outputs.
- Built and maintained JMeter and NeoLoad performance scripts for document upload, AI content extraction, and retrieval APIs; ran load tests with 100–300 concurrent users in AWS/PVDI environments.
- Monitored CloudWatch dashboards, identified latency issues and error-rate spikes, and partnered with developers on root-cause fixes.
- Delivered execution reports, KPI dashboards, and compliance documentation on time across 6 sprint cycles through weekly stakeholder meetings.

Senior Performance & Automation Engineer

Sep 2022 – Mar 2025

US Software Group

- Set NFRs and test strategy for Agthenticate (KDHI), a logistics identity-verification platform for driver and shipment authentication; built Selenium WebDriver automation packs with 60+ regression test cases for biometric enrollment/verification.
- Automated API testing for biometric login and shipment-tracking services (Python Requests, Postman); tested RBAC, location-based search, and biometric login (fingerprint, face recognition, OCR).
- Built functional, automation, and API test plans for Wurth Canada's B2B e-commerce and order management platform; maintained Selenium regression packs with 90+ test cases and automated pricing/inventory/order-status API checks.
- Ran JMeter load tests with 100–400 concurrent users across engagements to validate speed and response-time targets under peak load; verified backend data accuracy (MySQL) between storefront and order records.
- Supported WGS's enterprise web application with smoke/regression automation, REST API validation, and cross-browser functional testing for release consistency.

Quality & Performance Engineer

Jun 2021 – Sep 2022

Infedis Infotech LLP

- Set NFRs and built functional, automation, API, and performance test plans for HidocDr, a cross-platform (iOS/Android) medical learning application; authored 200+ test cases.
- Automated REST API testing for consultation-booking and CME endpoints (Postman, Python Requests); ran JMeter scenarios with 100–200 concurrent users to measure speed, latency, and session stability under peak load.
- Supported NFR gathering and test planning for Adacel's aviation training and simulation platform; tested integration APIs between simulation and training-record modules and verified backend data accuracy (MySQL).

Quality & Automation Engineer

Jun 2018 – Jun 2021

IDmission Solution Pvt. Ltd.

- Delivered functional, regression, API, database, and automation QA for a digital identity-verification SaaS platform; built Selenium WebDriver packs with 150+ regression test cases, cutting manual regression time by 2 days per release.
- Automated API testing for ID-verification and biometric login services (Python Requests, Postman); validated biometric login and backend data accuracy (MySQL) across multiple client integrations.
- Supported test strategy for a customer identity-verification and digital onboarding platform used in KYC workflows for a global banking client; ran JMeter load tests with 100–400 concurrent users and validated RBAC/location-based compliance checks.

CERTIFICATIONS

- Performance / Load Testing with JMeter and BlazeMeter
- Load Testing using NeoLoad (Valid up to 2026)

EDUCATION

- B.Tech / B.E. in Engineering — Nagpur University, 2016
- 12th (HSC) — Maharashtra Board, 2011
- 10th (SSC) — Maharashtra Board, 2009

LANGUAGES

English, Hindi, Marathi